PH1201 End Semester Examination

Full marks : 50 **Time : 2 hrs 30 minutes**

Name : **Roll No. :**

Attempt all questions

Q 1) a) An electric field is given by $\vec{E}(x,y,z) = x^3\hat{i} + (x-y^2)\hat{j} + \frac{xy}{z}\hat{k}$. At a point with coordinates $(2,1,3)$, find the local flux per unit area if the local area vector is $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)dS$.

b) Consider a hemisphere of radius R with its center at the origin and its curved surface is in $z \geq 0$. For an electric field $\vec{E} = \frac{E_0}{\sqrt{2}}\left(\hat{j}+\hat{k}\right)$ where E_0 is a constant, what is the electric flux through **(i)** the flat surface and **(ii)** the curved surface of the hemisphere? *Clearly indicate which normal you are using for the surface.* $[2+(3+3)]$

Q 2) a) A charge q is placed at the center of a unit cube. **(i)** Find the flux through any one of the faces of the cube. **(ii)** What is the net flux through any pair of opposite faces?

b) Find the r dependence of a spherically symmetric static charge density, $\rho(r)$, such that the magnitude of the corresponding electric field is r independent. $[(1+2)+5]$

Q 3) a) A particle of mass m and charge q is moving in the XY plane along the positive y axis with a velocity $\vec{v} = v_0\hat{j}$ when it encounters a uniform field $\vec{B} = B_0\hat{k}$ confined to $y > 0$. It enters the magnetic field at $(0,0,0)$ and travels in a curved path, finally returning to the field free region $(y < 0)$. **(i)** What are the coordinates of the point at which the particle exits the magnetic field? **(ii)** How much time did the particle spend in the magnetic field? **(iii)** What is its velocity after it exits the magnetic field?

b) Three concentric current carrying loops of radii R, $2R$ and $3R$ lie in the XY plane. The net magnetic field produced by them at their common center is zero. What is the current in the middle loop if the other two carry clockwise (as seen from above) currents, each equal to I?

$[(2+2+2)+3]$

Q 4) A square loop of side a has its center fixed at a distance D from an infinite wire co-planar with it. The wire carries a current I.

a) Calculate the flux of the magnetic field through the loop.

Siddhartha Lal

square

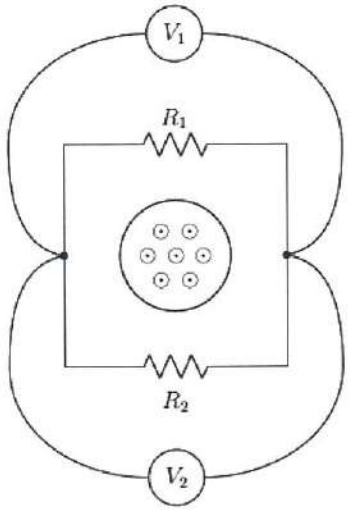
b) If the resistance of the ~~circular~~ loop is R, determine the net charge that flows through the loop if the current in the wire increases slowly (so that the quasi-static approximation can be used) from an initially steady value of I_0 to $2I_0$.

c) Sketch a diagram of the setup and indicate the direction of flow of the charge. You have to provide a justification for your answer. [3 + 3 + 2]

Q 5) a) Show that the four Maxwell equations are covariant under the parity transformation.

b) If charge is assumed to be a pseudoscalar rather than a scalar, the electric and magnetic fields must be pseudovector and vector, respectively. Show that even under this assumption the magnetic field due to an infinite straight wire would be tangential. [4 + 4]

Q 6) a) In the circuit shown, the infinite solenoid at the center (seen here in cross section) is being driven by a current of amplitude 50 mA and frequency of 50 Hz. The readings of the two ac voltmeters are 5 mV each when the two resistors are both equal to 1kΩ. We now change the resistor R_2 to 5 kΩ (keeping R_1 unchanged) while doubling the amplitude of the current driving the solenoid and increasing the frequency to 60 Hz. What do the two multimeters read now?



b) Starting from Maxwell's equations derive the wave equations in the presence of sources $\left(\rho \neq 0,\ \vec{j} \neq 0\right)$.

[5 + 4]

Useful Formulas / Results

The Maxwell equations in vacuum are

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \qquad \nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$
$$\nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Lorentz force law $\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$

Under a parity operation $\vec{r} \to -\vec{r}$, a vector changes sign, a pseudovector does not.

The magnetic field at the center of a circular current carrying loop is $\frac{\mu_0 I}{2r}$ directed normal to the loop.

The magnetic field due to an infinite current carrying wire at a distance d has a magnitude $\frac{\mu_0 I}{2\pi d}$.

Some useful vector identities:

$\nabla(fg) = f(\nabla g) + g(\nabla f)$
$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$
$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$
$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$

$\nabla \cdot (\nabla \times \mathbf{A}) = 0$
$\nabla \times (\nabla f) = \mathbf{0}$
$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$$
$$\int (\nabla \cdot \mathbf{A})\, d\tau = \oint \mathbf{A} \cdot d\mathbf{a}$$
$$\int (\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l}$$

VECTOR DERIVATIVES

Cartesian $\quad d\mathbf{l} = dx\,\hat{\mathbf{x}} + dy\,\hat{\mathbf{y}} + dz\,\hat{\mathbf{z}}, \quad d\tau = dx\,dy\,dz$

Gradient: $$\nabla t = \frac{\partial t}{\partial x}\hat{\mathbf{x}} + \frac{\partial t}{\partial y}\hat{\mathbf{y}} + \frac{\partial t}{\partial z}\hat{\mathbf{z}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Curl: $$\nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\mathbf{z}}$$

Laplacian: $$\nabla^2 t = \frac{\partial^2 t}{\partial x^2} + \frac{\partial^2 t}{\partial y^2} + \frac{\partial^2 t}{\partial z^2}$$

Spherical $\quad d\mathbf{l} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}, \quad d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi$

Gradient: $$\nabla t = \frac{\partial t}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial t}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial t}{\partial \phi}\hat{\boldsymbol{\phi}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}$$

Curl: $$\nabla \times \mathbf{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, v_\phi) - \frac{\partial v_\theta}{\partial \phi}\right]\hat{\mathbf{r}}$$
$$+ \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r}(r v_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{\boldsymbol{\phi}}$$

Laplacian: $$\nabla^2 t = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial t}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 t}{\partial \phi^2}$$

Cylindrical $\quad d\mathbf{l} = ds\,\hat{\mathbf{s}} + s\,d\phi\,\hat{\boldsymbol{\phi}} + dz\,\hat{\mathbf{z}}, \quad d\tau = s\,ds\,d\phi\,dz$

Gradient: $$\nabla t = \frac{\partial t}{\partial s}\hat{\mathbf{s}} + \frac{1}{s}\frac{\partial t}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial t}{\partial z}\hat{\mathbf{z}}$$

Divergence: $$\nabla \cdot \mathbf{v} = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

Curl: $$\nabla \times \mathbf{v} = \left[\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right]\hat{\mathbf{s}} + \left[\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right]\hat{\boldsymbol{\phi}}$$
$$+ \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_\phi) - \frac{\partial v_s}{\partial \phi}\right]\hat{\mathbf{z}}$$

Laplacian: $$\nabla^2 t = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial t}{\partial s}\right) + \frac{1}{s^2}\frac{\partial^2 t}{\partial \phi^2} + \frac{\partial^2 t}{\partial z^2}$$

Indian Institute of Science Education and Research Kolkata

Department of Physical Sciences

PH2201 : Basic Quantum Mechanics
End-semester Examination (Spring 2024)
Marks: 50, Duration: 2 hour and 30 min

1. A 24 watt LED light source having an area 50 cm^2 emits yellow light of wavelength 590 nm. As per Planck's proposal, how many light quanta come out of the light source per cm^2 area per second? Given $h = 6.6 \times 10^{-34}$ J $\cdot$ s and $c = 3 \times 10^8$ m $\cdot$ s^{-1}. [3]

2. Find the expectation values $\langle \hat{x} \rangle$ and $\langle \hat{p} \rangle$ in the state $|\psi\rangle = \frac{1}{2}(|1;t\rangle + |2;t\rangle)$ that describes a quantum SHO. [4+4]

3. If a state $|\psi\rangle$ of a quantum SHO gets annihilated by the operator $\hat{a}$ then find its normalized functional form in the p-representation $\psi(p) \equiv \langle p|\psi\rangle$. [6]

4. For a given operator $\hat{O} = -i\frac{d^2}{dx^2}$, find the corresponding adjoint operator $\hat{O}^\dagger$. Comment on the domain of the operator $\hat{O}^\dagger$. [3+2]

5. $\hat{L}_i$ with $i = x, y, z$ are self-adjoint angular momentum operators. (a) Show that the commutator bracket $[\hat{L}_z, \hat{L}^2] = 0$ where $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$. (b) Compute the commutator bracket $[\hat{L}_z, \hat{L}_+]$ where $\hat{L}_+ = i\hat{L}_x + \hat{L}_y$. [3+3]

6. In a Hydrogen atom, the radial wave-function $\psi = \psi(r)$ for an electron satisfies the equation

$$\frac{d^2(r\psi)}{d\rho^2} - \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2}\right](r\psi) = 0 \ ,$$

where $\rho = r\sqrt{-2mE/\hbar^2}$. Find the radial wave-function with total angular momentum $\sqrt{2}\hbar$ near the origin and far away from the origin. Sketch the probability of finding the electron along radial direction in both these regions. [5+3]

7. (a) Find the time-independent bound states for a particle of mass m which is moving in a potential well given by

$$V(x) = \begin{cases} 0 & |x| > 2a \\ -\frac{1}{2}V_0 & |x| \leq 2a \ . \end{cases}$$

(b) Determine the approximate energy eigenvalues, say E, for even bound states where $|2E + V_0|$ is small.
(c) Show that there exists at least one bound state even if the potential well is extremely shallow *i.e.* $V_0 \to 0$, and find the corresponding energy eigenvalue. [6+4+4]

G. M. Hossain

Koushik Dutta

PH2202 Thermal Physics
Fall Semester - 2024
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

End-semester Examination

Date: 2/05/2024

ALL THE BEST

1. (a) Define first and second-order phase transitions according to Ehrenfest's classifications. **Marks: 2**

   (b) Give one example for each case of first and second-order phase transitions. **Marks: 2**

2. Total internal energy of radiation in volume V is given by $U = AT^4V$, and pressure is $(1/3)AT^4$, where T is the temperature, and A is a constant. Find the entropy of the system. Using these results, show that the chemical potential for radiation is zero. **Marks: 3+2**

3. A system is changed from an initial state i to the final state f. Show that $(S_f - S_i) > \int_i^f \frac{dQ_{irr}}{T}$ where S_f is the final entropy and S_i is the initial entropy of the system. Here, dQ_{irr} is the differential heat transfer along an irreversible path with an instantaneous temperature T. **Marks: 3**

4. (a) An ideal one-mole gas of volume V_1 at temperature T is allowed to expand adiabatically to volume V_2 while doing no external work. The process was carried out abruptly, i.e. it was not a quasistatic process. Find out the final pressure of the gas. Each step of the argument/calculations must be written clearly. **Marks: 2**

   (b) For the above-mentioned process, calculate the entropy change between the final state and the initial state. Explain the steps clearly. **Marks: 2**

   (c) Calculate the entropy change again between the final state and the initial state of part (a) through a path other than the path taken in part (b) to argue that entropy is a state function. **Marks: 4**

5. Consider a gas system consisting of two subsystems separated by a wall that can exchange energy (not particles), and the wall is movable. The volume and the internal energy of the whole system remain constant. Find the condition for thermal and mechanical equilibrium between the two subsystems. **Marks: 4**

6. An ideal gas is contained in a large jar of volume V_0. Fitted to the jar is a glass tube of cross-sectional area A in which a metal ball of mass M just fits. The equilibrium pressure of the jar is slightly higher than the atmospheric pressure p_0 because of the weight of the ball. If the ball is displaced slightly from equilibrium, it will execute simple harmonic motion (neglect friction). If the state of the gas represents a quasistatic adiabatic process and γ is the adiabatic exponent, find oscillation frequency f in terms of the variables given in the problem. **Marks: 5**

7. Consider a gas in one spatial dimension of length L. Derive the density of states $D(\epsilon)$ as a function of energy ϵ of gas molecules. Note that $D(\epsilon)$ corresponds to number of states available between energy ϵ and $\epsilon + d\epsilon$. Take h to be the size of unit cell in the phase space. **Marks: 2**

8. Consider a system with three non-degenerate energy states with energies $\epsilon_1 = 0$, $\epsilon_2 = +\epsilon$ and $\epsilon_3 = +2\epsilon$. Two identical quantum particles are distributed in these states.

(a) Count the number of possible states when the particles are bosons and when they are fermions. It is wise to draw cartoon pictures. For each possible configuration, also note the total energy of the system. **Marks: 2+2**

(b) Write down the partition function for the *system* kept at a temperature T, when the particles are (i) bosons and (ii) fermions. **Marks: 2+1**

9. Consider a system of N particles each having two non-degenerate energy levels $+\epsilon$ and $-\epsilon$. The system is in thermal equilibrium at temperature T.

(a) Write down the partition function Z_1 for a single particle. **Marks: 2**

(b) Find out the Helmholtz free energy and the entropy of the system where the partition function for the system $Z_N = (Z_1)^N$. **Marks: 3**

(c) Show that the heat capacity at constant volume is given by

$$C_V = \frac{N\epsilon^2}{k_B T^2 \cosh^2(\epsilon/k_B T)}, \tag{1}$$

where k_B is the Boltzmann constant. Note that $\cosh\, x = \frac{1}{2}(e^x + e^{-x})$. It is suggested that you keep checking dimensions at each step of the calculations. **Marks: 5**

(d) From the above results make a qualitative plot of C_V/k_B (y-axis) vs $k_B T/\epsilon$ (x-axis). Pay special attention to $k_B T/\epsilon \ll 1$ and $k_B T/\epsilon \gg 1$ limit. **Marks: 2**

Huhhhu :-)

29/2

PH3201 End Semester Examination

Attempt all questions. Full Marks 50

1. Consider a function $f(x) = \begin{cases} k(x^2+x) & \text{if } 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$

(a) Find the value of k such that $f(x)$ can be considered as a probability distribution function.

(b) Hence find the cumulative distribution function $F(x)$.

(c) Draw a schematic sketch of $f(x)$ and $F(x)$, appropriately marking/labelling the axis. 3+3+2+2=10

2. The Hamiltonian of N non-interacting particles in equilibrium at temperature T is given by

$$H(q_1, q_2, \cdots, q_N, p_1, p_2, \cdots, p_N) \equiv a\ q_1^2 + \tilde{H}(q_2, \cdots, q_N, p_1, p_2, \cdots, p_N)$$

Write the canonical partition function and then compute $< aq_1^2 >$. 4+6= 10

3. A system has three energy levels of energy $-\epsilon,\ 0,\ +\epsilon$. As a function of temperature, T, find

(a) the partition function

(b) the average energy at $T = 0$ and $T = \infty$

(c) the entropy at $T = 0$ and $T = \infty$ 2+4+4=10

4. For $3N$ quantum harmonic oscillators at temperature T, where the energy level of a single oscillators is

$$\epsilon_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad n : 0,1,2,\cdots$$

(a) write the partition function of $3N$ non-interacting quantum harmonic oscillators

(b) find the average energy

(c) determine the heat capacity at constant volume

(d) show that in the temperature $\to \infty$ limit, Dulong Petit's law can be retrieved. 2+3+3+2=10

5. Consider a gas of electrons in a magnetic field, such that electrons can be either $\uparrow$ or $\downarrow$. In the limit of high temperature and vanishing magnetic field show that magnetic susceptibility obeys Curie's law. 10

End Semester Exam

PH3202, Spring 2024 | Instructor: Dr. Rajesh Kumble Nayak | DPS, IISER Kolkata

10:00 AM to 12:30 PM , 11th May 2024.

Duration 120 Min.

Rajesh

- Write neatly, and keep the rough work on the answer script. No writing on the question paper.
- Read the question carefully.
- Answer all questions.
- Write answers of all sub-parts of question in one place.
- Good luck!

Q - 1(20 Marks):

A long conducting cylinder with inner radius a and outer radius b carries a current I. The current density in the conductor is uniform. Find the magnetic field as a function of radius,

1. Inside the hollow region ($0 < r < a$).
2. Within the conductor ($a < r < b$).
3. Outside the conductor ($b < r$).

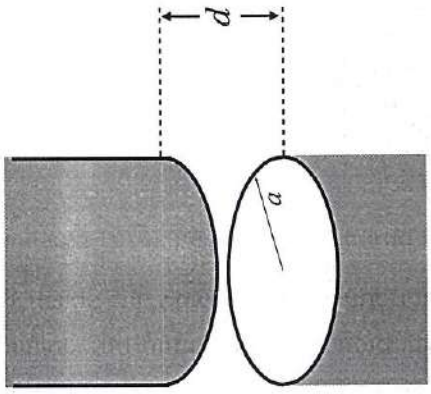
Q - 2(20 Marks):

A very long solenoid of radius b has n turns per unit length and carries a current $I = I_0 \sin \omega t$.

1. Find the magnetic field $\vec{B}$ inside the Solenoid as a function of time.
2. Find the electric field $\vec{E}$ inside and outside the solenoid as function of time.
3. Make a plot of $|\vec{E}|$ as function of radial coordinate at time $t = \frac{2\pi}{\omega}$.

Q - 3(20 Marks):

There a gap of distance d in cylindrical conductor of circular cross-section with a radius a (assume $d << a$), as shown in the figure,

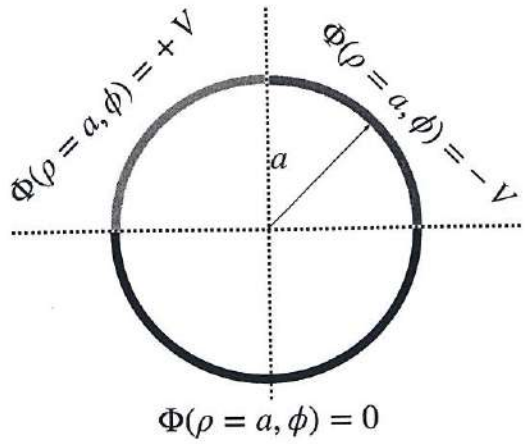


If a potential $\Phi = V_0 \cos \omega t$ is applied across the gap

1. Use the Maxwell's equations and symmetry arguments to determine Electric and Magnetic inside the gap.
2. If the current flow through the cylinder, find the current.

Q - 4(20 Marks):

A cylindrical plate with different sections is kept at different potentials, and the cross-section is shown in the figure below.

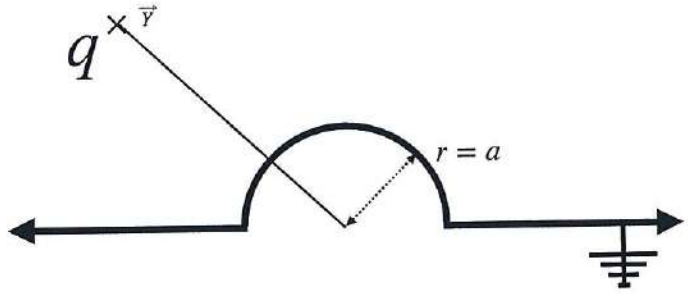


The boundary conditions are explicitly given by

$$\Phi(rho = a, \phi) = \begin{cases} -V & 0 < \phi < \pi/2 \\ +V & \pi/2 < \phi < \pi \\ 0 & \pi < \phi < 2\pi \end{cases}$$

1. Ignoring the z-axis, find the electrostatic potential in the interior region.
2. Find the induced charge density on the plate.

Q - 5 (20 Marks):



An infinity long ground conducting plate is embossed with a metallic half sphere in in the centre as shown in the figure. A point charge q is placed at an arbitrary position $\vec{Y}$ as shown in the figure.

- Find the potential in the region above the plate, using the method of imaging.
- Give one point justification for each of the image charges.

Q - 6 (20 Marks):

The Laplacian in a certain coordinate system $\{\sigma, \tau, \phi\}$ is given by

$$\nabla^2\Phi = \frac{1}{\sigma^2 + \tau^2}\left[\frac{1}{\sigma}\frac{\partial}{\partial\sigma}\left(\sigma\frac{\partial\Phi}{\partial\sigma}\right) + \frac{1}{\tau}\frac{\partial}{\partial\tau}\left(\tau\frac{\partial\Phi}{\partial\tau}\right)\right] + \frac{1}{\sigma^2\tau^2}\frac{\partial^2\Phi}{\partial\phi^2} = 0.$$

Separate the above Laplacian in to 3 ODE's

PH3203: Advanced Quantum Mechanics

End Semester Examination

3rd May, 2024

Total Marks: 100 **Time - 2 hours 30 minutes**

Instructions: *Write your full name and Roll no. clearly on both the question paper and the answer script. Mark the correct option/options in the question paper. Each correct option in the multiple-choice questions will carry equal weight. Marking any incorrect option will lead to zero credit. Provide the full calculations for multiple-choice questions also. Please note that for each question, 1 mark will be deducted for messy, untidy and disorganized answers.*

Name ..... **Roll No.** .....

Q.1) The scattering matrix corresponding to a localized potential (positioned at $x = 0$) in one dimension is given by $S = e^{i\alpha \mathbb{I}} e^{i\vec{B}\cdot\vec{\sigma}}$, (where $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$).

(i) Show that S is unitary.

(ii) Obtain the condition on $\alpha, \vec{B}$ such that

(a) S respects time-reversal symmetry.

(b) S respects parity symmetry.

(c) S respects both time reversal and parity symmetry.

(iii) Obtain the probability current density for an incident wave with momentum $p = \hbar k$.

(iv) Assuming $B_z = 0$, calculate the transfer matrix corresponding to this scattering matrix, S.

Note : Show the precise steps to obtain the results.

Marks : 3+(2+2+2)+4+7 = 20

Q.2) Consider a particle of mass m which is initially ($t < 0$) trapped in the ground state of a 1D infinite potential box of length L.

$$\begin{aligned} V(x) &= 0, & -L/2 < x < L/2 \\ &= \infty, & \text{elsewhere} \end{aligned}$$

Assume that the particle is subjected to the following time-dependent perturbation,

$$\begin{aligned} V_p(t) &= -q\mathcal{E}x, & 0 \le t \le \tau \\ &= 0, & t > \tau \end{aligned}$$

Lets denote the transition probability from i-th level to f-th level (obtained from k-th order perturbation theory) as $P^{(k)}_{i\to f}$.

(i) For first-order perturbation theory,

(A) $P^{(1)}_{1\to 2} = 0$

(B) $P^{(1)}_{1\to 3} = 0$

(C) $P^{(1)}_{1\to 2} = 4(q^2\mathcal{E}^2)\left(\frac{16L}{9\pi^2}\right)^2\left(\frac{2mL^2}{3\pi^2\hbar^2}\right)^2\sin^2\left(\frac{3\pi^2\hbar\tau}{4mL^2}\right)$

(D) $P^{(1)}_{1\to 3} = 9(q^2\mathcal{E}^2)\left(\frac{32L}{27\pi^2}\right)^2\left(\frac{2mL^2}{3\pi^2\hbar^2}\right)^2\sin^2\left(\frac{8\pi^2\hbar\tau}{9mL^2}\right)$

(ii) For second-order perturbation theory,

(A) $P^{(2)}_{1\to 2} = 0$

(B) $P^{(2)}_{1\to 3} = \left|\sum_{n=\text{even}} \frac{192q^2\mathcal{E}^2(2m^2L^3)^2}{(n^2-1)^3(n^2-9)^2\pi^6\hbar^2}\left[\frac{\left(e^{i\frac{4\pi^2\hbar\tau}{mL^2}}-1\right)}{8\pi^2\hbar^2} - \frac{\left(e^{i\frac{(9-n^2)\pi^2\hbar\tau}{2mL^2}}-1\right)}{(9-n^2)\pi^2\hbar^2}\right]\right|^2$

(C) $P^{(2)}_{1\to 3} = \left|\sum_{n=\text{even}} \frac{96q^2\mathcal{E}^2(2m^2L^3)^2}{(n^2-1)^3(n^2-9)^2\pi^6\hbar^2}\left[\frac{\left(e^{i\frac{4\pi^2\hbar\tau}{mL^2}}-1\right)}{8\pi^2\hbar^2} - \frac{\left(e^{i\frac{(9-n^2)\pi^2\hbar\tau}{2mL^2}}-1\right)}{(9-n^2)\pi^2\hbar^2}\right]\right|^2$

(D) $P^{(2)}_{1\to 3} = 0$

Marks : 5+10 = 15

Given : The n-th eigenstate for the infinite well potential is given by

$$\psi_n(x) = \sqrt{\frac{2}{L}}\cos\frac{n\pi x}{L} \qquad n = 1, 3, 5, \cdots\cdots$$
$$= \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L} \qquad n = 2, 4, 6, \cdots\cdots$$

$$P^{(1)}_{i\to n} = \left|\left(-\frac{i}{\hbar}\right)\int_0^\tau dt\, e^{i(E_n-E_i)t/\hbar}V_{ni}(t)\right|^2$$
$$P^{(2)}_{i\to n} = \left|\left(-\frac{i}{\hbar}\right)^2\sum_m\int_0^\tau dt\, e^{i(E_n-E_m)t/\hbar}V_{nm}(t)\int_0^{t'} dt'\, e^{i(E_m-E_i)t'/\hbar}V_{mi}(t')\right|^2$$

Q.3) Consider a particle, moving in one dimension, subjected to the potential,

$$V(x) = -V_0, \qquad x \geq 0$$
$$= 0, \qquad \text{elsewhere}$$

where $V_0 > 0$.

(i) If we write the trial wavefunction as $e^{ikx} + re^{-ikx}$ for $x < 0$ and $t\,e^{ik'x}$ for $x > 0$, then which of the following is true?

(A) $r + t = 1$

(B) $r - t = 1$

(C) $|r|^2 + |t|^2 = 1$

(D) $t - r = 1$

(ii) The phase shift in transmission amplitude with respect to the incident wave is

(A) 0

(B) π

(C) $\frac{\pi}{2}$

(D) $\tan^{-1}\left(\frac{k'}{k}\right)$

(iii) The correct expression for reflection probability (R) is given by,

(A) $\frac{(k-k')^2}{(k+k')^2}$

(B) $\frac{4kk'}{(k+k')^2}$

(C) $\frac{4k^2}{(k+k')^2}$

(D) $\frac{4k'^2}{(k+k')^2}$

Marks : 5+5+5 = 15

Q.4) Consider the following two level Hamiltonian $\hat{H} = E\sigma_z$ and at any time t the particle state can be represented as $|\psi(t)\rangle = [c_1(t) \quad c_2(t)]^T$. It is known that for $t < 0$, the particle is at the state with energy E. For $t \geq 0$, the particle is subjected to a time-dependent perturbation $\hat{V}(t)$, where

$$\hat{V}(t) = \left(\cos(\omega t)\sigma_x - \sin(\omega t)\sigma_y\right)\gamma \quad , \gamma = \text{constant}$$

Consider, $\omega_0 = -2E/\hbar$.

(i) The Schrodinger equation in the interaction picture can be written as,

(A) $i\hbar\dot{c}_1 = \gamma e^{i(\omega-\omega_0)t}c_2$ and $i\hbar\dot{c}_2 = \gamma e^{-i(\omega-\omega_0)t}c_1$

(B) $i\hbar\dot{c}_1 = \gamma e^{i(\omega-\omega_0)t}c_1$ and $i\hbar\dot{c}_2 = \gamma e^{-i(\omega-\omega_0)t}c_2$

(C) $i\hbar\dot{c}_1 = \gamma e^{-i(\omega-\omega_0)t}c_2$ and $i\hbar\dot{c}_2 = \gamma e^{i(\omega-\omega_0)t}c_1$

(D) $i\hbar\dot{c}_1 = \gamma e^{i(\omega+\omega_0)t}c_2$ and $i\hbar\dot{c}_2 = \gamma e^{-i(\omega+\omega_0)t}c_1$

(ii) The minimum time taken by the particle to return back to the initial state is given by,

(A) $\frac{\pi}{\left[\frac{\gamma^2}{\hbar^2}+\frac{(\omega-\omega_0)^2}{2}\right]^{1/2}}$

(B) $\frac{\pi}{2\left[\frac{\gamma^2}{\hbar^2}+\frac{(\omega-\omega_0)^2}{4}\right]^{1/2}}$

(C) $\frac{\pi}{\left[\frac{\gamma^2}{\hbar^2}+\frac{(\omega-\omega_0)^2}{4}\right]^{1/2}}$

(D) $\frac{2\pi}{\left[\frac{\gamma^2}{\hbar^2}+\frac{(\omega-\omega_0)^2}{4}\right]^{1/2}}$

Marks : 5 + 10 = 15

Q.5) Consider scattering from a 3D spherically symmetric potential of the form

$$\begin{aligned} V(r) &= V_0, \qquad & r \leq r_0 \\ &= 0, \qquad & r > r_0 \end{aligned}$$

Find the scattering phase shift in the s-wave channel for (i) $V_0 > 0$ and (ii) $V_0 < 0$.

Marks : 10

Hint : The radial part of the Schrodinger equation is given by

$$\Big[-\frac{d^2}{dr^2}+\frac{2m}{\hbar^2}V(r)\Big]\big(rR_{l=0}(r)\big)=k^2\big(rR_{l=0}(r)\big)$$

where $k^2=\frac{2mE}{\hbar^2}$ with $E>0$.

For $k^2>k_0^2$ (where $k_0^2=\frac{2mV_0}{\hbar^2}$), the trial solution is

$$\begin{aligned} rR_{l=0}(r) &= \sin\Big(\sqrt{k^2-k_0^2}\,r\Big), && r<r_0 \\ &= \sin(k\cancel{\phi}\,r+\delta_0), && r>r_0 \end{aligned}$$

Q.6) Consider a particle of mass m and energy E, which is incident from the left on the potential of the form

$$\begin{aligned} V(x) &= 0 && x<-a \\ &= -V_0 && -a\le x\le 0 \\ &= \infty && x>0 \end{aligned}$$

If the incoming wave is Ae^{ikx}, (assume $k=\sqrt{2mE}/\hbar$ and $k'=\sqrt{2m(E+V_0)}/\hbar$)

(i) Find the reflected wave.

(A) $B=Ae^{-ika}\left[\frac{k-ik'\cot k'a}{k+ik'\cot k'a}\right]$

(B) $B=Ae^{-ika}\left[\frac{k+ik'\cot k'a}{k-ik'\cot k'a}\right]$

(C) $B=Ae^{-2ika}\left[\frac{k-ik'\cot k'a}{k+ik'\cot k'a}\right]$

(D) $B=Ae^{-2ika}\left[\frac{k+ik'\cot k'a}{k-ik'\cot k'a}\right]$

(ii) Find the phase shift δ for a very deep well, (i.e. in the limit $E<<V_0$)

(A) $\delta=ka$

(B) $\delta=-ka$

(C) $\delta=2ka$

(D) $\delta=-2ka$

Marks : 10+5 = 15

Q.7) Consider a pair of spin - 1/2 particles emitted by a local source such that the first particle is received by observer A and the second particle is received by observer B. Assuming that these pair of particles is prepared in the state :

$$|\psi\rangle=\frac{1}{\sqrt{3}}\big(|\uparrow^z\rangle_1\otimes|\downarrow^z\rangle_2+|\uparrow^z\rangle_1\otimes|\uparrow^z\rangle_2+|\downarrow^z\rangle_1\otimes|\downarrow^z\rangle_2\big),$$

(i) obtain the joint probability $p(A,B|+1,-1)$ of σ_x with measurement outcome $+1$ for the first particle and $\sigma_{\hat{n}}=\cos\theta\sigma_x+\sin\theta\sigma_y$ with measurement outcome -1 for the second particle.

(ii) Evaluate the correlation function

$$E(A,B)=\sum_{a,b=\pm1}ab\,p(A,B|a,b)$$

$A=\sigma_x\,;\,B=\sigma_{\hat{n}}$

Marks : 4+6 = 10

Indian Institute of Science Education and Research, Kolkata
Department of Physical Sciences
Advanced Stat Physics (PH4202)
Instructor: *P. K. Mohanty*

04/05/2024

Endsem Exam (4th May, 2024, 14:00-16:30) **Full Mark: 30**

Attempt six questions from {1(a),1(b),1(c),2,3,4,5,6,7 }. Each carry [5] marks.

Notations: $\beta = \frac{1}{k_B T}$, T is temperature and k_B is Boltzman constant.

1. Consider 3-state Potts model on a a periodic one dimensional lattice of size $2L$, where each site $i = 1, 2, \ldots, 2L$ with energy function

$$E(\{s_i\}) = \sum_{i=1}^{2L} [\mu - J\delta_{s_i, s_{i+1}}], \tag{1}$$

where the spin s_i at site i takes the values $s_i = 0, 1, 2$. The partition function of the system can be written as $Z_{2L} = Tr\left[M^{2L}\right]$, where the transfer matrix $M \equiv M(x, y)$ with $y = e^{-\beta\mu}$ and $x = e^{\beta J}$.
(a) Use Migdal-Kadanoff decimation method to show that, under RG transformation,

$$x \to x' = \frac{2 + x^2}{1 + 2x} \quad \text{and} \quad y \to y' = \frac{y^2}{1 + 2x}$$

Hint: Integrate even-spins $\{s_2, s_4, \ldots, s_{2L}\}$, get new partition function $Z_L = Tr[(M(x', y'))^L]$. **[5]**
(b) Show that $x^* = 1$ is an attractive fixed point of this transformation $x' = \frac{2+x^2}{1+2x}$. This fixed point corresponds to temperature $T \to \infty$ or $J \to 0$, and thus, long range order is not possible for any finite T. Also show that at any fixed $T > 0$, if one starts with $J < 0$ (anti-ferromagnetic interaction) the systems generates an effective ferromagnetic interaction ($J' > 0$) in one RG-step, and it remains ferromagnetic in subsequent RG steps. **[5]**
(c) For a q-sate Potts model, with spins $s_i = 0, 1, \ldots, q-1$ and $E(\{s_i\})$ given by Eq. (1), show that that the transfer matrix can be written as $yM(x)$ with $M(x) = |v\rangle\langle v| + (x-1)I_q$. Here, I_q is the identity matrix in $q-$ dimension, and the vector $\langle v| = (1, 1, \ldots)$. Find how x and y transforms under decimation [Find x' and y' as a function of (x, y).] **[5]**

2. For a $L \times L$ periodic lattice partition function is $Z = Tr[M^{L^2}]$. Two competing eigenvalues of the transfer matrix $\lambda_1(\beta)$ and $\lambda_2(\beta)$ are given below; all other eigenvalues are much smaller than $\lambda_{1,2}(\beta)$. Decide in each case, if the system can exhibit a ~~first-order, a second-order or no~~ phase transition in the thermodynamic limit.
(i) $\lambda_1(\beta) = 12e^{\beta} + e^{3\beta}$, $\lambda_2(\beta) = 6e^{2\beta} + 8$ (ii) $\lambda_1(\beta) = 12e^{\beta} + e^{2\beta} + 6$, $\lambda_2(\beta) = 10e^{2\beta} + 1$
(iii) $\lambda_1(\beta) = 12e^{\beta} + e^{2\beta}$, $\lambda_2(\beta) = 10e^{2\beta} + 8$
For those undergo a phase transition, find T_c and draw a schematic plot of free energy.
Hint: Take $x = e^{\beta}$ and find x_c. **[5]**

3. The sites of a $L \times L \times L$ cubic lattice, periodic in all there directions, are defined by (x, y, z) where $x, y, z = 0, 1, \ldots, L-1$.
(a) Convert these L^3 points to one dimensional array indexed by a unique integer $k = 0, 1, \ldots, L^2 - 1$. Find a function $k = f(x, y, z)$ that serves the purpose. **[1]**
(b) Each site (x, y, z) has six neighbours, index them as $m = (0, 1, 2)$ when the neighbours are along +ve (x, y, z) directions, and $m = (3, 4, 5)$ when the neighbours are along −ve (x, y, z) directions respectively. Find the functions $g_m(x, y, z)$ that returns the k-value corresponding to m-th neighbour (x, y, z).
Hint: Use $mod(m, n)$ to represent remainder of m after division with n. **[4]**

4. An one-dimensional lattice gas [periodic L-size lattice with site variables $n_i = 0$ (empty), 1 (occupied by one hard-core particle)] is described by an energy function

$$E(\{n_i\}) = \sum_{i=1}^{L} (n_i + 2n_{i+1} + 3n_{i+2}).$$

When the system is attached to a bath at temperature T and chemical potential μ, the total number of particles N fluctuates; what is its distribution function $P(N)$? **[5]**

5. Partition function of an one-dimensional periodic system (sites $i = 1, 2 \ldots, L$) is $Tr[M^L]$. Let eigenvalues of M are ordered as $\lambda_{max} < \lambda_1 < \lambda_2 \cdots < \lambda_L$. For any local observable A_i, show that the correlation function $G(r) = \langle A_i A_{i+r} \rangle - \langle A_i \rangle \langle A_{i+r} \rangle$ can be written as [5]

$$G(r) \sim e^{-r/\xi} \text{ where } \xi^{-1} = \ln \frac{\lambda_1}{\lambda_{max}}.$$

Hint: $\langle A_i \rangle = \frac{Tr[M^i \hat{A} M^{L-i}]}{Tr[M^L]}$ where $\hat{A}$ is a constant matrix corresponding to the local observable A_i.

6. Near the critical point the orderparameter ϕ of a system is given by

$$\phi(\epsilon, h) = \int_0^\infty q(x, \epsilon, h) dx$$

where the function $q(x, \epsilon, h)$ has a scaling form $q(x, \epsilon) = x^{-a} g(\frac{\epsilon}{x^b}, \frac{h}{x^c})$, and $g(.,.)$ scales as $g(x, y) = x^{-m} f(x/y^n)$. Given that $f(.)$ has the following property: for large x, $f(x) \sim x^s$; and for small x $f(x)$ approach a constant $f_0 > 0$. Express exponents s, m, n in terms of standard critical exponents β and γ defined by, [5]

$$\phi(\epsilon, h) \sim \begin{cases} \epsilon^\beta & \text{when } h = 0 \\ h^{\frac{1}{\gamma}} & \text{when } \epsilon = 0 \end{cases}.$$

7. The Gaussian model in 3-dimension is defined by the following free energy functional

$$F[\phi(\vec{x})] = \frac{1}{2} \int d\vec{x} \{ r\phi^2(\vec{x}) + c(\nabla \phi(\vec{x}))^2 \}.$$

From the exact solution, find the critical point and show that the correlation length of the system diverges there with a critical exponent $\nu = \frac{1}{2}$. [5]

End-semester examination: PH4204 (2024)

Time : 2 hours 30 minutes : Full marks = 50

Answer all questions.

1. Starting from definitions, show that the $SU(3)$ group has 8 generators.

[8]

2. Starting from definitions, obtain how the following quantities written in terms of the electric and magnetic fields transform under charge conjugation and parity:

   (a) $\vec{E} \times \vec{B}$

   (b) $\vec{\nabla}(\vec{E}.\vec{B})$

[4+4]

3. Suppose, $f_u^v(p)$ and $f_d^v(n)$ are, respectively, the values of the valence up-quark distribution in a proton, and the valence down-quark distribution in a neutron, for any given value of the momentum fraction x.

   (a) How are $f_u^v(p)$ and $f_d^v(n)$ related in the lowest order?

   (b) Which symmetry of strong interaction is responsible for your conclusion?

[2+2]

4. Let there be a scalar field ϕ and two spin-1/2 fermion fields f_1, f_2, with masses m_1, m_2. They interact via the following Lagrangian:

   $\mathcal{L} = y\bar{f}_1\gamma_5 f_2\phi \; + \; h.c.$

   (a) Write down, without neglecting any masses, the lowest order transition amplitude (leaving out powers of i) for the decay $\phi(k) \to f_1(p_1)\bar{f}_2(p_2)$, where the corresponding four-momenta are given within brackets.
   (b) Obtain the modulus square of the amplitude, summed over the polarisation states of each fermion.

[3+~~5~~ 6]

5. (a) Draw the properly labelled lowest order Feynman diagram according to the Glashow-Salam-Weinberg theory, for the weak decay

$$\mu^-(k) \to \nu_\mu(p_1)e^-(p_2)\bar{\nu}_e(p_3)$$

(b) Write down the corresponding amplitude (upto factors of i), in the gauge where the W-boson propagator with four-momentum q is $\frac{-g_{\mu\nu}+q_\mu q_\nu/m_W^2}{q^2-m_W^2}$

[5+4]

6. (a) Taking into account *only electromagnetic interaction*, draw the two lowest order Feynman diagrams corresponding to

$$e^-(p_1)e^+(p_2) \to e^-(p_3)e^+(p_4)$$

(b) What is the relative sign between the s-and t-channel diagram amplitudes? Justify your answer.

(c) Draw the additional diagrams that contribute to the same transition in the same order, if now weak interaction, too, is switched on.

[4+4+4]

IISER Kolkata
End-Semester Examination 2024, Semester VIII
PH 4205(General Relativity and Cosmology)
Answer all questions. Time: 2 hours 30 minutes; Full Marks:50

1. a) For the timelike velocity vector u^μ, the acceleration vector a^μ is defined as $a^\mu = u^\mu_{;\nu}u^\nu$. Show that the acceleration vector is orthogonal to the velocity vector. (4)
b) Show that by using a quadratic transformation of coordinates, $x'^\mu = x^\mu + \frac{1}{2}Q^\mu_{\alpha\beta}x^\alpha x^\beta$ where $Q^\mu_{\alpha\beta}$ are constants and symmetric in the lower indices, one can find out a "Locally Lorentz" coordinate system. The transformation of the Christoffel symbols are given as
$\Gamma'^j_{km} = \frac{\partial x'^j}{\partial x^i}\frac{\partial x^l}{\partial x'^k}\frac{\partial x^p}{\partial x'^m}\Gamma^i_{pl} + \frac{\partial^2 x^i}{\partial x'^k \partial x'^m}\frac{\partial x'^j}{\partial x^i}$. (6)

c)ξ^μ is a Killing vector field and $T^{\mu\nu}$ is a symmetric tensor satisfying the condition $T^{\mu\nu}_{;\nu} = 0$. Show that $A^\mu_{;\mu} = 0$ where $A^\mu = T^{\mu\nu}\xi_\nu$. (4)

2. a)Starting from the geodesic equation $\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$, show that for the weak field approximation ($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$) for a slowly moving particle, $h_{00} = 2\phi$ where ϕ is the Newtonian gravitational potential (i.e., show that $\ddot{x}^i = -\eta_{ij}\frac{\partial\phi}{\partial x^j}$). (6)

b) The Bianchi identity is given as $G^{\mu\nu}_{;\nu} = (R^{\mu\nu} - \frac{1}{2}Rg^{\mu\nu})_{;\nu} = 0$. Using Einstein,s equations, show that for a vacuum, the Bianchi identity looks like $R^{\mu\nu}_{;\nu} = 0$. (4)

3. The Schwarzchild spacetime is given by

$$ds^2 = (1 - \frac{2m}{r})dt^2 - (1 - \frac{2m}{r})^{-1}dr^2 - r^2 d\Omega^2.$$

a) Write down a suitable Lagrangian and set up the geodesic equations for a timelike particle. (6)
b) Show that if you start with the initial condition $\theta = \frac{\pi}{2}$ and $\dot{\theta} = 0$, the motion is restricted in the $\theta = \frac{\pi}{2}$ plane. (3)
c) Now using the first integrals of these equation in the $\theta = \frac{\pi}{2}$ plane, set up a differential equation for $u = \frac{1}{r}$ with ϕ as the argument. (5)

4. Einstein equations for a spatially flat, homogeneous and isotropic universe are given by

$$\frac{3\dot{a}^2}{a^2} = 8\pi G\rho,$$

$$\frac{2\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} = -8\pi G p,$$

where a, ρ, p are the scale factor, the density and pressure of the fluid respectively.
a) Find a as a function of the cosmic time (t) and ρ as a function of a for a radiation-filled universe ($p = \frac{\rho}{3}$). (5)

b) Establish a relation between the cosmic time t and the radiation temperature T. (4)

c) Define the deceleration parameter q in terms of a and its derivatives. Find the value of q for the spatially flat radiation dominated universe. (3)

PH4207 :: Quantum Information Processing
Final Examination (May 08, 2024)

Marks: 40, Duration: 2 hours 30 mins

1. (a) For a two qubit state,

$$|\Phi\rangle = \tfrac{1}{2}|\uparrow\rangle_A \left(\tfrac{1}{2}|\uparrow\rangle_B + \tfrac{\sqrt{3}}{2}|\downarrow\rangle_B\right) + \tfrac{1}{2}|\downarrow\rangle_A \left(\tfrac{\sqrt{3}}{2}|\uparrow\rangle_B + \tfrac{1}{2}|\downarrow\rangle_B\right).$$

Perform the the Schmidt decomposition of $|\Phi\rangle$. [5]

(b) Is there a Schmidt decomposition for an arbitrary *tripartite* pure state? That is, if $|\psi\rangle_{ABC}$ is an arbitrary vector in $H_A \bigotimes H_B \bigotimes H_C$, can we find orthonormal bases $\{|i_A\rangle\}, \{|i_B\rangle\}, \{|i_C\rangle\}$, such that $|\psi\rangle_{ABC} = \sum_i \lambda_i |i_A\rangle \otimes |i_B\rangle \otimes |i_C\rangle$? Explain your answer. [3]

2. Consider the two qubit state

$$\rho = p\,|\psi\rangle\langle\psi| + \frac{1-p}{4}\mathbf{I}$$

where $|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. Here $\boldsymbol{I}$ is the density operator on the 2-qubit Hilbert space.

(a) Having written down ρ in the computational basis, examine the entanglement of this mixed state density matrix using Peres Horodecki partial transpose criterion. [3]

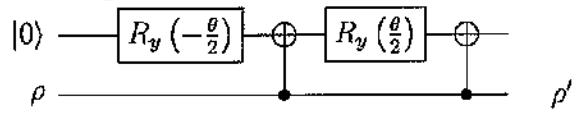
(b) Find the range of values of p for which the state is entangled. [1]

(c) Discuss the positivity (whether a positive map or a completely positive map) of partial transposition operation. [3]

(d) How can we say that the set of separable density matrices form a convex set. Using this property comment on the validity of Peres Horodecki partial transpose criterion to determine entanglement. [3]

3. Draw the quantum circuit of the Quantum Fourier Transform (QFT) operation on 3 qubits, explicitly explaining all the terms logically. Also, write down the matrix of this QFT in the computational basis and show that the QFT is a unitary operation. Compute explicitly the QFT of the 3 qubit state $\{|101\rangle\}$. Outline Shor's algorithm in your own language. State why QFT is considered important in Shor's algorithm? [10]

4. (a) Given the Kraus operator $F_0 = \begin{bmatrix} 1 & 0 \\ 0 & cos\theta \end{bmatrix}$ & $F_1 = \begin{bmatrix} 0 & 0 \\ 0 & sin\theta \end{bmatrix}$, consider the quantum circuit shown in the figure below



Show that this circuit induces a quantum operation $\rho \to \rho' = F_0 \rho F_0^\dagger + F_1 \rho F_1^\dagger$ [4]

(b) Consider a single qubit (spin qubit) state governed by the Hamiltonian,

$$H = -\frac{\omega_0}{2}\sigma_z$$

where ω_0 is the energy difference between the ground state $|0\rangle$ and the excited state $|1\rangle$. Write down with justification the Lindblad operator with appropriate constants for spontaneous emission from excited state to ground state. Write down the Lindblad equation for this system, and solve for the full density matrix. Provide the full solution in terms of time dependence of all the components of the density matrix. Assume appropriate boundary conditions.

[8]

30/4/24

Date : 30th April, 2024

End-semester Exam

PH4208 : Quantum Field Theory 2

Duration: 150 minutes

Weight: 50 %

1. **Dirac matrices properties:** Consider Dirac equation and matrices γ^μ in $(3+1)$-dimensions and prove the following identities.

   (a) $Tr\left(\gamma^\mu\gamma^\nu\right) = 4\eta^{\mu\nu}$ and $Tr\left(\gamma^\mu\gamma^\nu\gamma^5\right) = 0$ 2

   (b) $\exp\left(i\alpha\gamma^5\right) = \cos(\alpha) + i\gamma^5\sin(\alpha)$ 4

   (c) $\exp\left(ip_\mu\gamma^\mu\right) = \cos(\cancel{p^2}) + i\frac{p_\mu\gamma^\mu}{p}\sin(p)$, where $p = \sqrt{p_\mu p^\mu}$. Assume p to be dimensionless. 4

   $\cos(p)$

2. **Yukawa theory:** Consider the following theory Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{m^2}{2}\phi^2 + \bar{\psi}(i\ \partial\!\!\!/ - M)\psi - iy\ \bar{\psi}\gamma^5\psi\phi - \frac{g}{6}\phi^3 - \frac{\lambda}{24}\phi^4,$$

   where ψ is a Dirac fermion and ϕ is a pseudo-scalar.

   (a) Write down the renormalized perturbation theory for the above Lagrangian along with the expressions for all the counter terms in general. 2

   (b) Assuming the parity to be a good symmetry at the quantum level, write down the renormalization conditions for the above theory. 3

   (c) If we denote the fermion ψ propagator by $S(x_1, x_2)$ and pseudo-scalar ϕ propagator by $D(x_1, x_2)$ then draw the following diagram:

$$D(x_1, z_1)S(z_1, z_2)S(z_2, z_3)S(z_3, z_1)D(z_2, z_3)$$

   and write the expression for it. Show that the diagram evaluates to zero using properties of γ matrices in $(3+1)$-dimensions. 3

   (d) Write down the *partition function* $Z_0[J, \eta, \bar{\eta}]$ for the free part of the above theory as a function of J, η and $\bar{\eta}$. 2

   (e) Write down the *partition function* $Z[J, \eta, \bar{\eta}]$ for the full theory in terms of functional derivatives of $Z[J, \eta, \bar{\eta}]$ and write down the full expression of the first order term $\omega_1(J, \eta, \bar{\eta})$. 10

3. **Linear sigma model with Yukawa:** Consider the following theory Lagrangian:

$$\frac{1}{2}(\partial_\mu\phi_i)(\partial^\mu\phi_i) + \frac{\mu^2}{2}\phi_i^2 - \frac{\lambda}{4}(\phi_i^2)^2 + \bar{\psi}(i\ \partial\!\!\!/)\psi - g\bar{\psi}(\phi_1 + i\gamma^5\phi_2)\psi, \qquad i = 1,\ 2.$$

   (a) Show that $\phi_1 \to \cos\alpha\ \phi_1 - \sin\alpha\ \phi_2$, $\phi_2 \to \sin\alpha\ \phi_1 + \cos\alpha\ \phi_2$ and $\psi \to \exp[-i\gamma^5\alpha/2]\psi$ is a symmetry of the theory. 2

   (b) Exand the Lagrangian near the minima $(\phi_1(x), \phi_2(x)) = (v + \sigma(x), \pi(x))$ and write all the interactions in terms of σ, ϕ and ψ. 2

   π

(c) Show that the superficial degree of divergence in d space-time dimensions is given by

$$\begin{aligned} D &= d - \left[\frac{d-2}{2}\right](N_\sigma + N_\pi) - \left[\frac{d-1}{2}\right] N_\psi \\ &+ \left[\frac{d-6}{2}\right](V_{\sigma^3} + V_{\sigma\pi^2}) + \cancel{\left[\frac{d-6}{2}\right](V_{\sigma^3} + V_{\sigma\pi^2})} \\ &+ \left[\frac{d-4}{2}\right](V_{\sigma\psi^2} + V_{\pi\psi^2}) + [d-4](V_{\sigma^2\pi^2} + V_{\pi^4} + V_{\sigma^4}) \end{aligned}$$

where N_P is the number of external legs in the Green's function of type P and V_I is the number of interaction vertex of type I. 4

(d) For $d = 4$, list all the Green's functions that are, *in principle*, divergent. 2

(e) Write down the renormalized perturbation theory in terms of σ, ϕ and ψ and Feynman rules for the counter terms. 4

(f) Consider the fermion loop correction to the one point function $\langle\Omega|\sigma(x)|\Omega\rangle$ at one loop. Show that this diagram's divergent part is of order $\mathcal{O}(g^4)$. 6

Useful Formula:

$$\int \frac{d^d l}{(2\pi)^d} \frac{1}{(l^2 - \Delta)^n} = \frac{(-1)^n}{(4\pi)^d/2} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-\frac{d}{2}}$$

$$\Gamma(x) = \frac{1}{x} - \gamma_E + \mathcal{O}(x)$$

Evolutionary Dynamics (PH4209) Final Exam (May 9, 2024)

Answer all questions. **Total Time=2 hours. Total Marks=30.** The marks are specified in brackets.

1.a) The following reactions lead to Rock-Paper-Scissors dynamics. Write down the set of dynamical equations for the frequencies ($x_A; x_B; x_C$) of A, B, C respectively. ϕ – empty site; p,q,z are the rates of the respective reactions (3)

$$AB \xrightarrow{p} A\emptyset, \quad BC \xrightarrow{p} B\emptyset, \quad CA \xrightarrow{p} C\emptyset,$$
$$AB \xrightarrow{z} AA, \quad BC \xrightarrow{z} BB, \quad CA \xrightarrow{z} CC,$$
$$A\emptyset \xrightarrow{q} AA, \quad B\emptyset \xrightarrow{q} BB, \quad C\emptyset \xrightarrow{q} CC.$$

b) If in addition to the above reactions, A, B, C can mutate to each other with a rate μ i.e. (3)

$$A \xrightarrow{\mu} \begin{cases} B \\ C \end{cases}, \quad B \xrightarrow{\mu} \begin{cases} A \\ C \end{cases}, \quad C \xrightarrow{\mu} \begin{cases} A \\ B \end{cases}$$

Write down the new set of dynamical equations for the frequencies $x_A; x_B; x_C$

2. In a game with two strategies A and B and a population of N players among whom i play strategy A and (N-i) play strategy B. The pay-off matrix elements E(A,A)=a, E(A,B)=b, E(B,A)=c, E(B,B)=d

a) Write down the total expected payoff to A and B respectively. (2)

b) If f_i and g_i be the fitness of A and B respectively when i players playing strategy A and evolution occurs according to a Moran process, write down the expressions of f_i and g_i. What is the probability that the number of players playing (i) strategy A increases by one (ii) strategy B increases by one? (2)

c) The condition that the fixation probability of strategy A is greater than the neutral fixation probability leads to the inequality **a(N-2) + b (2N-1) > c (N+1) + d(2N-4)**

If the payoffs f_i and g_i are equal, then show that the unstable equilibrium frequency (x*=i/N) of A-players in the N>>1 limit satisfies the constraint x*<1/3 by using the above inequality. (3)

3. (a) Imagine a structured population of size N described by a **Line** graph where a node can only be replaced by the offspring of a node preceding it and the last node in the linear chain replaces itself. Write down the N x N stochastic matrix that specifies the probability with each node can replace any other node in the graph. What is the probability of fixation of a mutant that arises in such a structured population and at which location should that mutant appear in order to be fixed in the population? Justify your answer! (2)

(b) In a **directed cycle** graph, every member can only be replaced in the next generation by the member *preceding* it. If the #B members are *i* and #A=*N-i*, what is the probability that the #B individuals (i) decrease by 1 and (ii) increase by 1? Justify your answer in each case. Assume fitness of B=r, fitness of A=1. (2+2)

(c) (i) Consider a PD game (with pay-off matrix elements E(A,A)=a, E(A,B)=b, E(B,A)=c, E(B,B)=d) on a regular network, where each node has k-connections. $q_{A|B}$ represents the probability that a B-player has a A-player as a neighbour etc. If a B-player is selected for death

Write down the payoff P_i to the ith-player (i=A-neighbour of the B player selected for death or B-neighbour of the B-player selected for death); explain each term in the expressions for the two payoffs. (2)

(ii) If a B-player is selected for death in the network, obtain an expression for the probability that the B-player is replaced by one of it's A-neighbours. Explain the significance of each term in the expression. (4)

4 (a) Inactivating a tumour suppressor gene (TSG) requires two mutations, one in each allele. The first mutation that inactivates one of the alleles occurs with the rate u1 and the second mutation that inactivates the remaining allele occurs with a rate u2. Type 0 cells have two unmutated alleles and Type 1 cells have only one mutated allele. If there are *i* cells of Type 1 and *(N-i)* cells of type 0, write down the probabilities for number of Type **1** cells to (i) increase by 1 (ii) remain unchanged. Briefly justify your answers.
(2+2)

(b) The N+2 state Markov process needed to describe TSG inactivation as described in part (a) reduces to a 3-state Markov process in the limit of $N << \frac{1}{\sqrt{u_2}}$.

(i) Define the 3 states by clearly explaining what each of them represent. (2)

(ii) Write down the dynamical equations for $X_0(t)$, $X_1(t)$ and $X_2(t)$ which describe the rate of change of these quantities over time. $X_0(t)$, $X_1(t)$ and $X_2(t)$ correspond to the probability of finding the system in each of the 3 possible states. Explain each of the terms appearing in the equations for $X_0(t)$, $X_1(t)$ and $X_2(t)$. (3)

9/8/24

PH4213 End Semester Examination

Full marks : 50 **Time : 2 hrs 30 minutes**

Attempt any five questions

Q 1) a) Using geometrical arguments find the vector $\vec{r}'$ that is obtained when a vector $\vec{r}$ is rotated (actively) through an angle θ about an axis $\hat{n}$.

b) Two successive active rotations, through $45°$ each about the X, and Y axes, respectively are carried out. Find

i) the angle of the resulting rotation

ii) its axis. $[4 + (2 + 4)]$

Q 2) a) Show that the group manifold for $SU(2)$ is the sphere S^3.

b) (i) Determine the composition law for the Poincaré group.

(ii) Hence show that the Lorentz and translation groups are subgroups of the Poincaré group. Which one of these is a normal subgroup and why? $[4 + (2 + 4)]$

Q 3) a) Starting from the associative property of a Lie group derive the relation

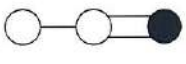
$$\frac{\partial f^j}{\partial \beta^k}(\alpha;\beta) = \hat{\eta}^j{}_p\left(f(\alpha;\beta)\right)\hat{\xi}^p{}_k(\beta)$$

where $\hat{\eta}^j{}_p(\alpha) = \frac{\partial f^j}{\partial \beta^k}(\alpha;\beta)\Big|_{\beta=0}$ and the $\hat{\xi}$s are defined by $\hat{\eta}^j{}_p(\alpha)\,\hat{\xi}^p{}_k(\alpha) = \delta^j_k$.

b) Derive the two compatibility conditions that follow from the above once you use the demand that the mixed partial derivatives $\frac{\partial^2 f^j}{\partial \beta^l \partial \beta^k}$ and $\frac{\partial^2 f^j}{\partial \beta^k \partial \beta^l}$ are equal. Show that these two compatibility conditions are mutually equivalent. $[3 + (4 + 3)]$

Q 4) Consider the Lie algebra spanned by two elements a and b, with the relation $[a, b] = b$. Is this Lie algebra solvable? Is it nilpotent?

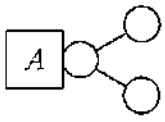
b) Show that the set of all upper triangular $n \times n$ matrices form a Lie algebra. Prove that the derivative subalgebra of this Lie algebra is the set of all strictly upper triangular $n \times n$ matrices. $[4 + (2 + 4)]$

Q 5) a) Write down the Cartan matrix corresponding to Dynkin diagram  . *Note : we use the convention where shorter root is represented by a filled circle.*

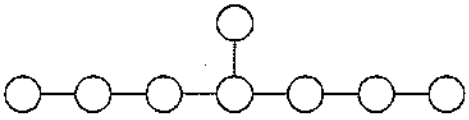
Ananda Dasgupta

b) Determine all the roots of the Lie algebra represented by this diagram.

c) Determine the weights of all states in the representations with Dynkin factors $(2, 0, 0)$. $[2 + 4 + 4]$

Q 6) a) Prove that if the structure  leads to a valid Dynkin diagram then the structure is also a valid Dynkin diagram.

b) Show that the structure



can not be a part of a valid Dynkin diagram. $[4 + 6]$